

FCFS CPU Scheduling

Aim

To write and execute a Python program to implement the First-Come, First-Served (FCFS) CPU scheduling algorithm and calculate the waiting time and turnaround time for each process, along with their averages.

Program

```
GNU nano 8.7.1                                fcfs.py *
def find_wt(process, n, bt, at):
    wt = [0] * n
    service_time = [0] * n
    service_time[0] = at[0]

    for i in range(1, n):
        service_time[i] = service_time[i - 1] + bt[i - 1]
        wt[i] = service_time[i] - at[i]

        if wt[i] < 0:
            wt[i] = 0
            service_time[i] = at[i]

    return wt

def find_TAT(n, bt, wt):
    tat = [0] * n

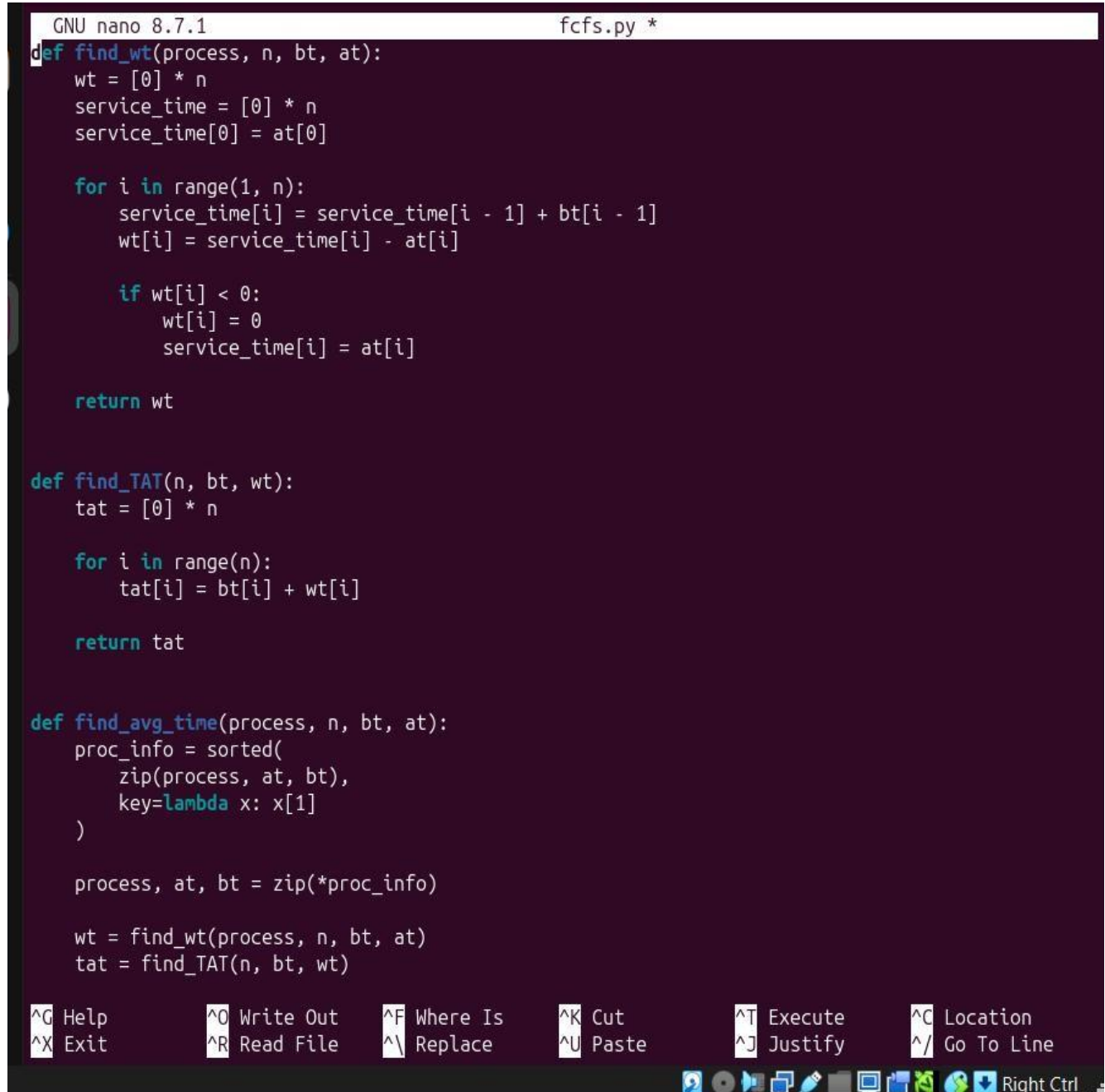
    for i in range(n):
        tat[i] = bt[i] + wt[i]

    return tat

def find_avg_time(process, n, bt, at):
    proc_info = sorted(
        zip(process, at, bt),
        key=lambda x: x[1]
    )

    process, at, bt = zip(*proc_info)

    wt = find_wt(process, n, bt, at)
    tat = find_TAT(n, bt, wt)
```



```
total_wt = sum(wt)
total_tat = sum(tat)

print("Execution order:", " -> ".join(map(str, process)))

print("\nProcess\tArrival\tBurst\tWaiting\tTurnaround")

Settings i in range(n):
    print(
        f"{process[i]}\t{at[i]}\t{bt[i]}"
        f"\t{wt[i]}\t{tat[i]}"
    )

print(f"\nAverage Waiting Time = {total_wt / n:.2f}")
print(f"Average Turnaround Time = {total_tat / n:.2f}")

if __name__ == "__main__":
    process = [1, 2, 3, 4]
    arrival_time = [0, 1, 2, 3]
    burst_time = [5, 3, 8, 6]

    find_avg_time(
        process,
        len(process),
        burst_time,
        arrival_time
    )
```

Help Write Out Where Is Cut Execute Location
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Output :

Output for the given process, arrival-time, and burst-time values:

```
python3: can't open file '/home/vansh/fcfs.py': [Errno 2] No such file or directory
vansh@ubuntu:~$ nano fcfs.py
vansh@ubuntu:~$ python3 fcfs.py
Execution order: 1 -> 2 -> 3 -> 4

Process Arrival Burst   Waiting Turnaround
1         0         5         0         5
2         1         3         4         7
3         2         8         6        14
4         3         6        13        19

Average Waiting Time = 5.75
Average Turnaround Time = 11.25
vansh@ubuntu:~$
```

Result

The FCFS scheduling program was executed. The average waiting time is 5.75 time units and the average turnaround time is 11.25 time units.

SJF CPU Scheduling

Aim

To write and execute a Python program to implement the Shortest Job First (SJF) CPU scheduling algorithm and calculate the waiting time and turnaround time for each process, along with their averages.

Program

```
GNU nano 8.7.1 sjf.py *
def find_waiting_time(process, n, bt, wt):
    wt[0] = 0

    for i in range(1, n):
        wt[i] = bt[i - 1] + wt[i - 1]

def find_turnaround_time(n, bt, wt, tat):
    for i in range(n):
        tat[i] = bt[i] + wt[i]

def find_avg_time(process, n, bt):
    wt = [0] * n
    tat = [0] * n

    # Sort processes according to burst time
    for i in range(n):
        for j in range(i + 1, n):
            if bt[i] > bt[j]:
                bt[i], bt[j] = bt[j], bt[i]
                process[i], process[j] = process[j], process[i]

    find_waiting_time(process, n, bt, wt)
    find_turnaround_time(n, bt, wt, tat)

    total_wt = sum(wt)
    total_tat = sum(tat)

    print("SJF Execution Order:")
    print(" -> ".join(map(str, process)))

    print("\nProcess\tBurst Time\tWaiting Time\tTurnaround Time")

    for i in range(n):
        print(
            f"{process[i]}\t{bt[i]}\t\t"
            f"{wt[i]}\t\t{tat[i]}"
        )

    print(f"\nAverage Waiting Time = {total_wt / n:.2f}")
    print(f"Average Turnaround Time = {total_tat / n:.2f}")

if __name__ == "__main__":
    process = [1, 2, 3, 4]
    burst_time = [5, 3, 8, 6]

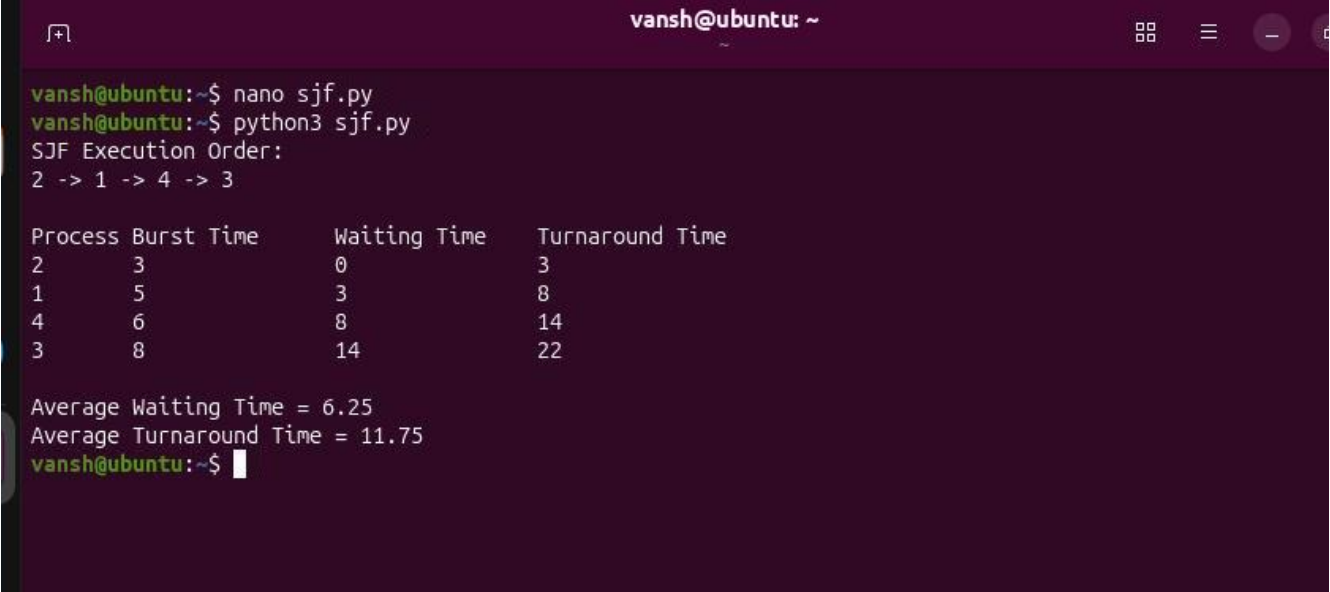
    find_avg_time(
        process,
        len(process),
        burst_time
    )
```

GNU nano 8.7.1 sjf.py *

Help Exit Write Out Read File Where Is Replace Cut Paste Execute Justify Location Go To Line Undo Redo Set Mark Copy To Bracket Where Was Right Ct

Output Screenshot

Output for the given burst-time values:



```
vansh@ubuntu: ~  
vansh@ubuntu:~$ nano sjf.py  
vansh@ubuntu:~$ python3 sjf.py  
SJF Execution Order:  
2 -> 1 -> 4 -> 3  
  
Process Burst Time      Waiting Time      Turnaround Time  
2         3             0                 3  
1         5             3                 8  
4         6             8                14  
3         8            14                22  
  
Average Waiting Time = 6.25  
Average Turnaround Time = 11.75  
vansh@ubuntu:~$
```

The screenshot shows a terminal window with a dark purple background. The user 'vansh' is logged into an 'ubuntu' machine. They run 'nano sjf.py' to edit a file and then 'python3 sjf.py' to execute it. The program outputs the SJF execution order as '2 -> 1 -> 4 -> 3'. It then displays a table with four columns: 'Process', 'Burst Time', 'Waiting Time', and 'Turnaround Time'. The data rows are: Process 2 with burst 3, wait 0, turnaround 3; Process 1 with burst 5, wait 3, turnaround 8; Process 4 with burst 6, wait 8, turnaround 14; and Process 3 with burst 8, wait 14, turnaround 22. Below the table, it calculates the 'Average Waiting Time = 6.25' and 'Average Turnaround Time = 11.75'. The prompt 'vansh@ubuntu:~\$' is visible at the bottom.

Result

The SJF scheduling program was executed successfully. The average waiting time is 6.25 time units and the average turnaround time is 11.75 time units.